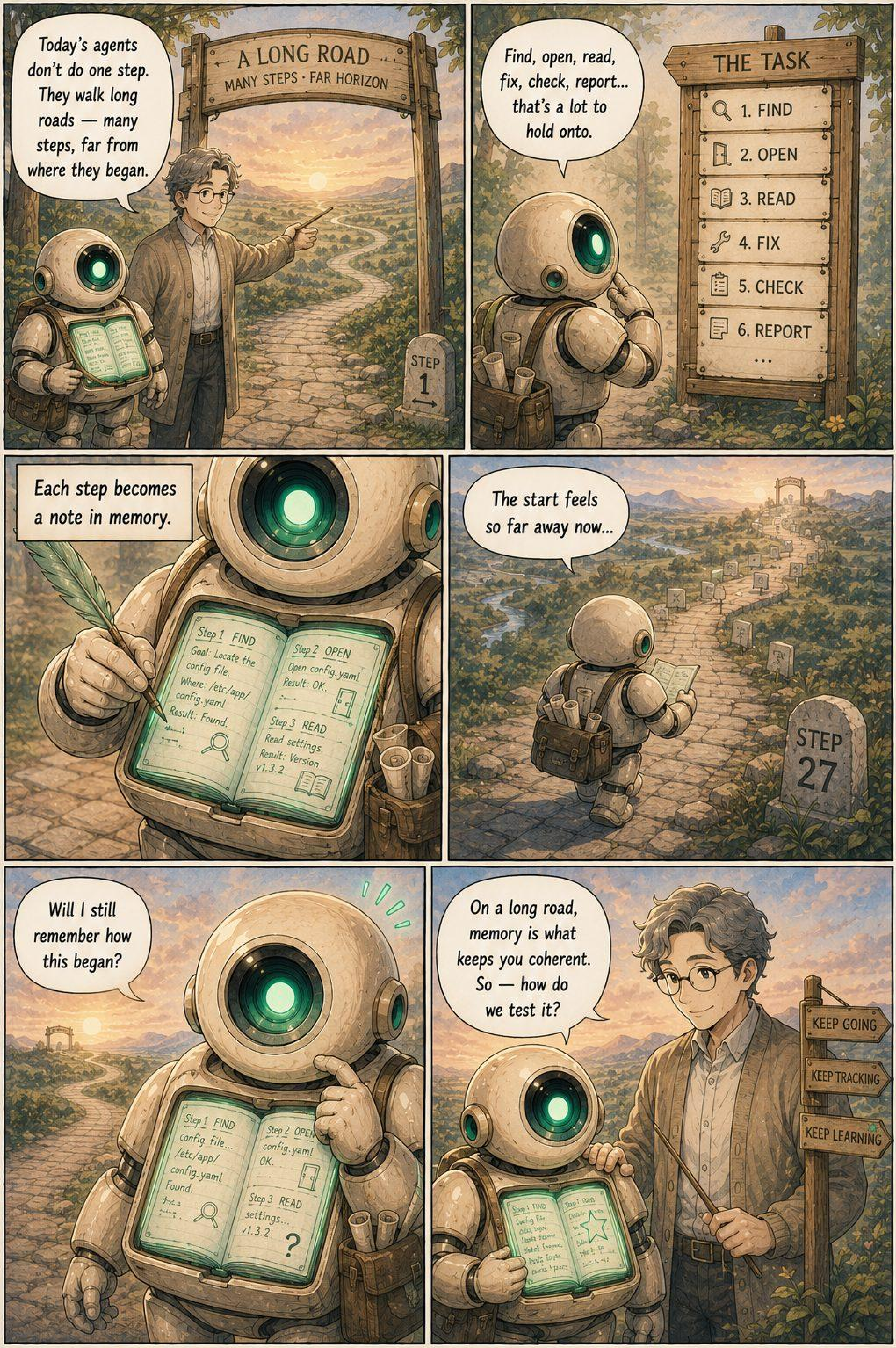
ENGRAM//KIT

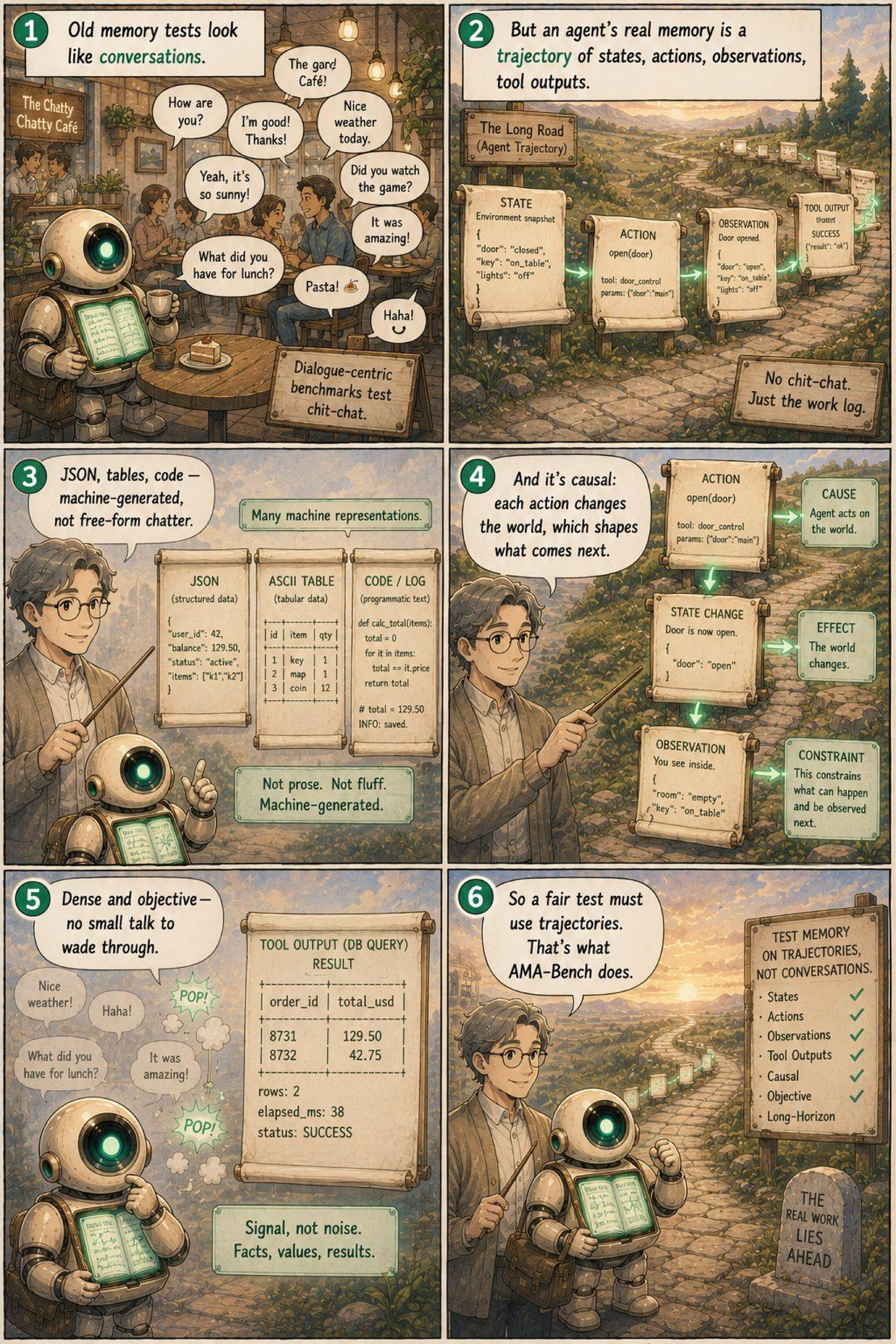
educational manga reading pack

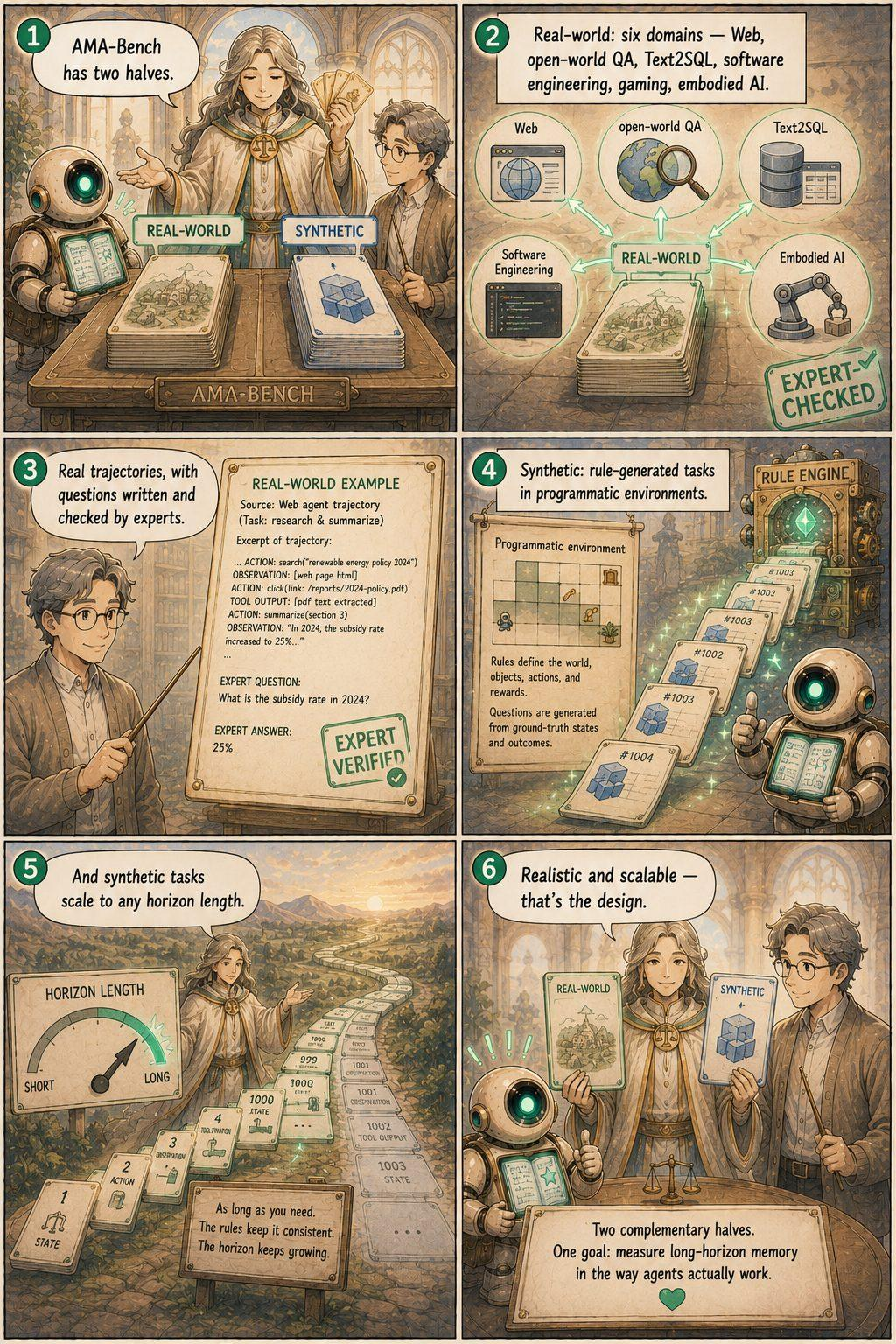
The Long Road Test

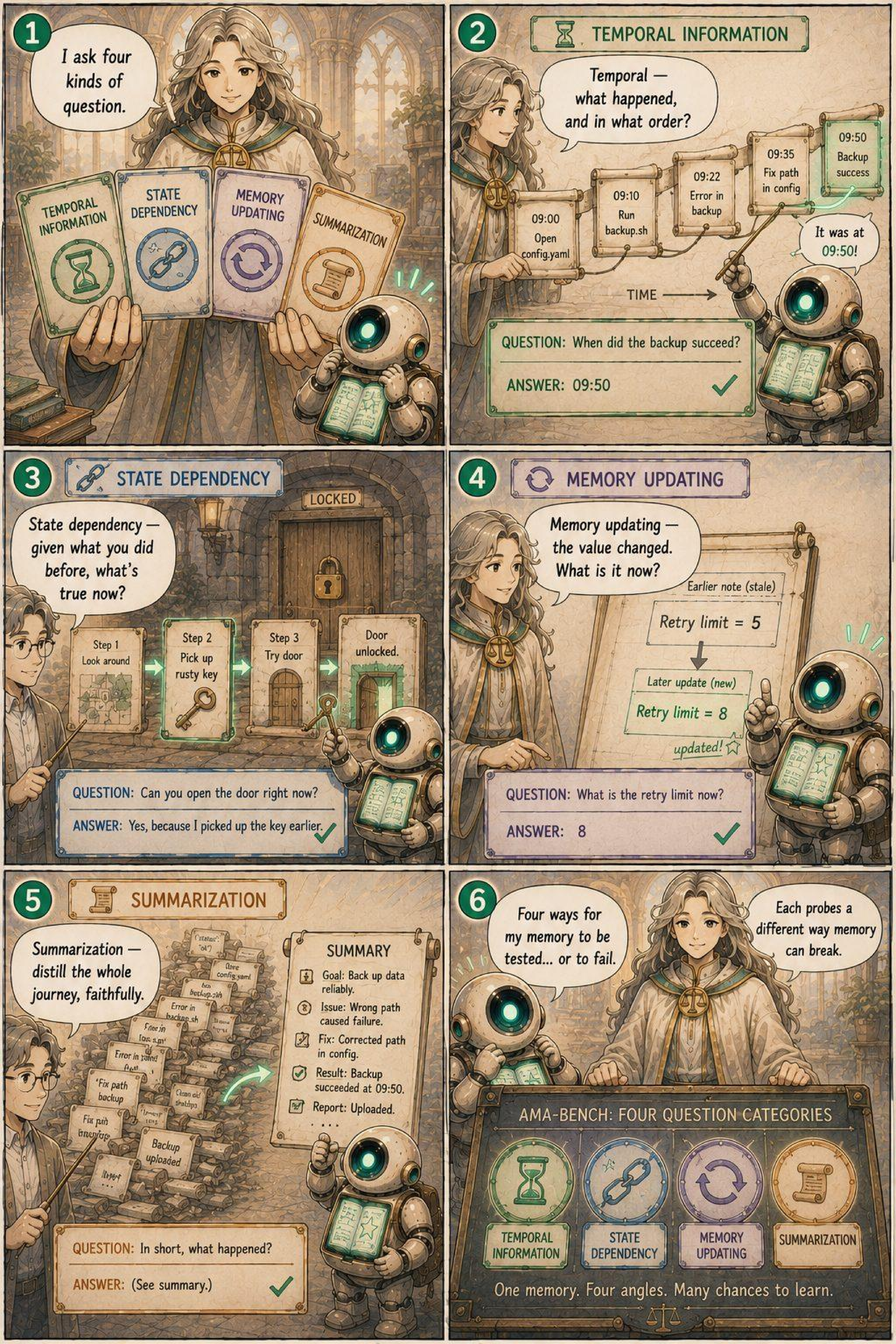
AMA-Bench

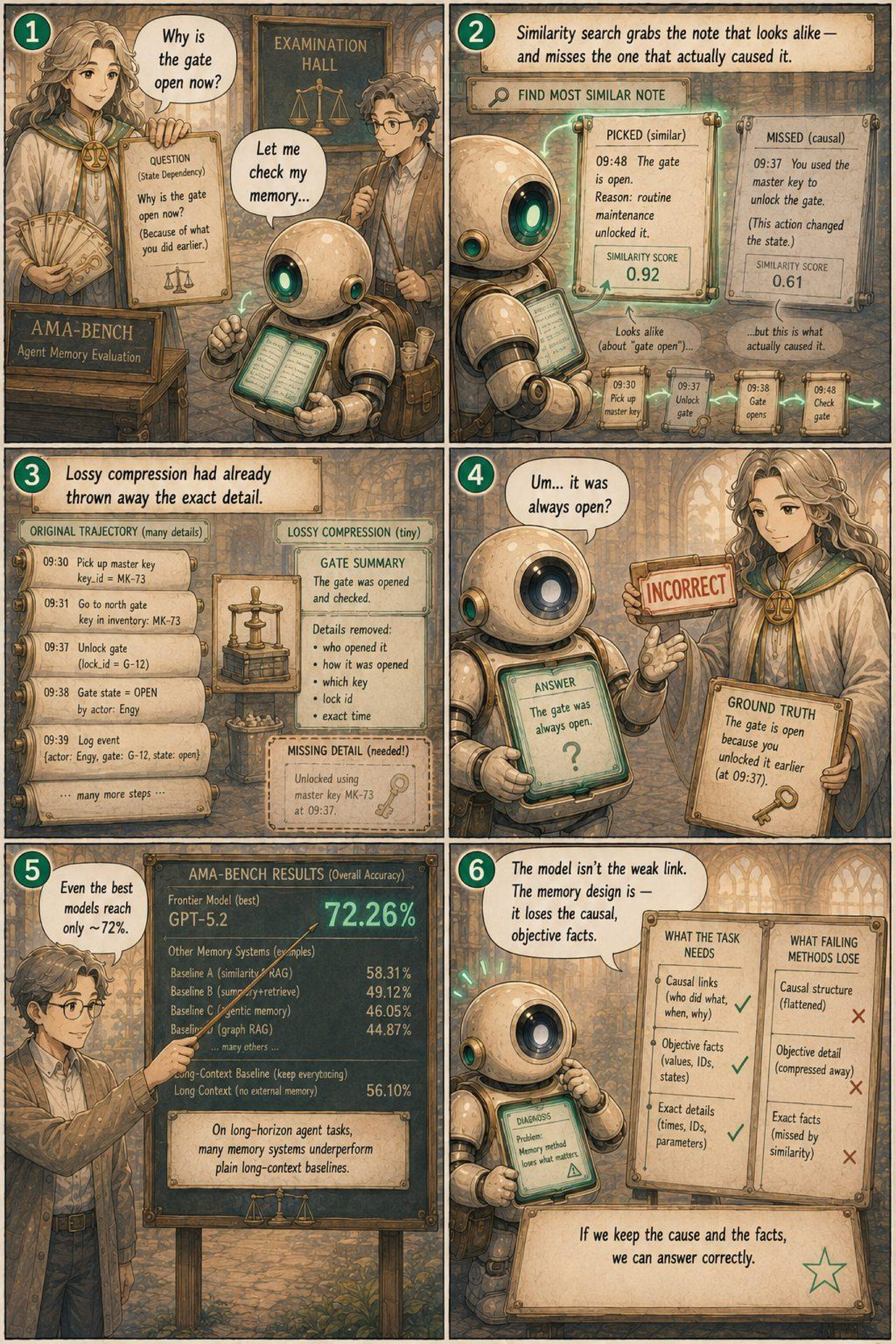
github.com/romyilano/engram-kit


A LONG ROAD
MANY STEPS · FAR HORIZON
Today's agents don't do one step. They walk long roads — many steps, far from where they began.
STEP 1
Find, open, read, fix, check, report... that's a lot to hold onto.
THE TASK
1. FIND
2. OPEN
3. READ
4. FIX
5. CHECK
6. REPORT
...
Each step becomes a note in memory.
Step 1 FIND
Goal: Locate the config file.
Where: /etc/app/config.yaml
Result: Found.
Step 2 OPEN
Open config.yaml
Result: OK.
Step 3 READ
Read settings.
Result: Version v1.3.2
The start feels so far away now...
STEP 27
Will I still remember how this began?
Step 1 FIND
config file...
/etc/app/config.yaml
Found.
Step 2 OPEN
config.yaml
OK.
Step 3 READ
settings...
v1.3.2
?
On a long road, memory is what keeps you coherent. So — how do we test it?
KEEP GOING
KEEP TRACKING
KEEP LEARNING

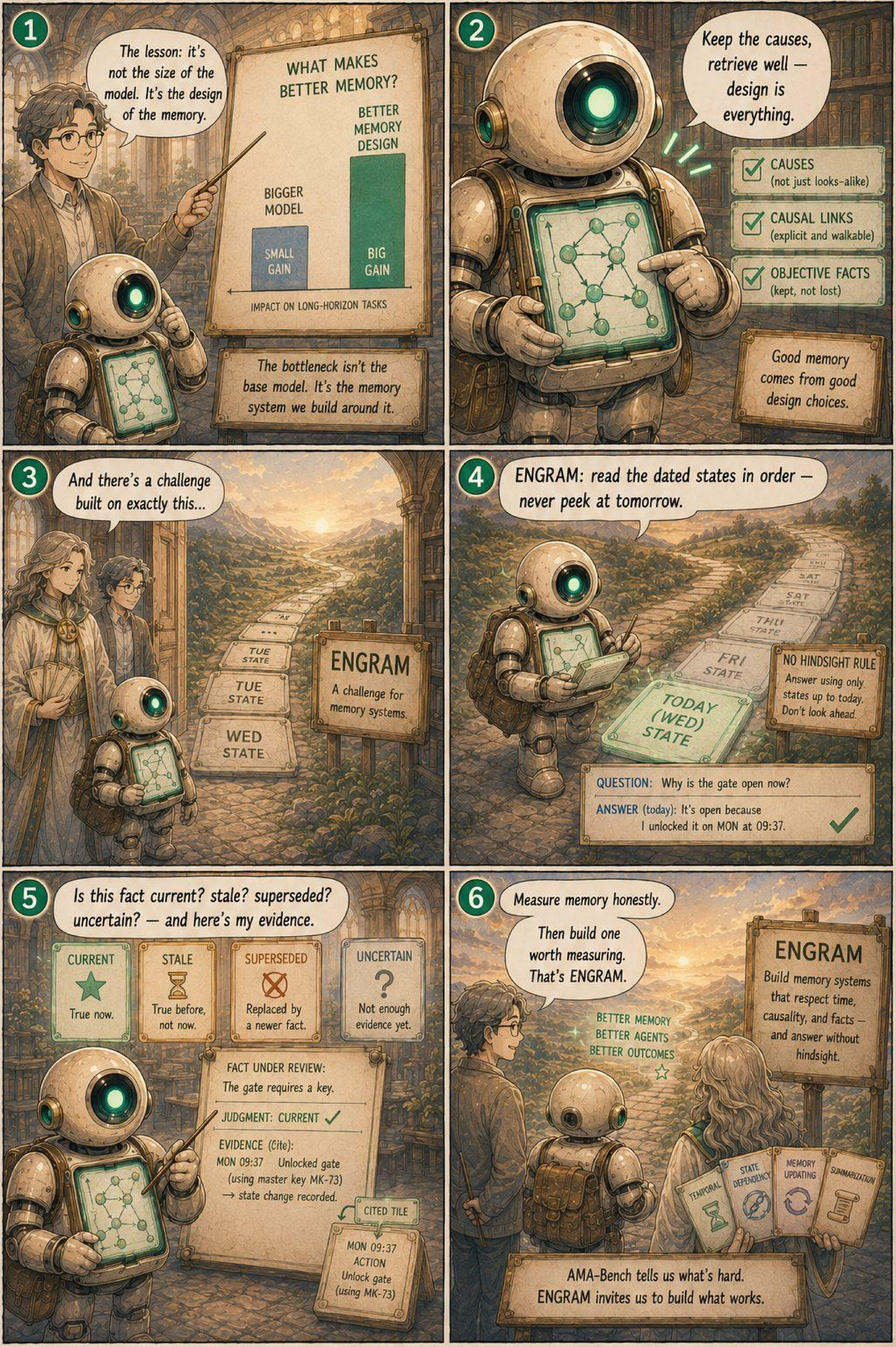

1
Old memory tests look like conversations.
The Chatty Chatty Café
How are you?
The gard Café!
I'm good! Thanks!
Nice weather today.
Yeah, it's so sunny!
Did you watch the game?
It was amazing!
What did you have for lunch?
Pasta!
Haha!
Dialogue-centric benchmarks test chit-chat.
2
But an agent's real memory is a trajectory of states, actions, observations, tool outputs.
The Long Road (Agent Trajectory)
STATE
Environment snapshot
{ "door": "closed", "key": "on_table", "lights": "off" }
ACTION
open(door)
tool: door_control
params: {"door":"main"}
OBSERVATION
Door opened.
{ "door": "open", "key": "on_table", "lights": "off" }
TOOL OUTPUT
SUCCESS
{"result": "ok"}
No chit-chat. Just the work log.
3
JSON, tables, code — machine-generated, not free-form chatter.
Many machine representations.
JSON
(structured data)
{ "user_id": 42, "balance": 129.50, "status": "active", "items": ["k1","k2"] }
ASCII TABLE
(tabular data)
id	item	qty
1	key	1
2	map	1
3	coin	12
CODE / LOG
(programmatic text)
def calc_total(items):
total = 0
for it in items:
total += it.price
return total
total = 129.50
INFO: saved.
Not prose. Not fluff. Machine-generated.
4
And it's causal: each action changes the world, which shapes what comes next.
ACTION
open(door)
tool: door_control
params: {"door":"main"}
CAUSE
Agent acts on the world.
STATE CHANGE
Door is now open.
{ "door": "open" }
EFFECT
The world changes.
OBSERVATION
You see inside.
{ "room": "empty", "key": "on_table" }
CONSTRAINT
This constrains what can happen and be observed next.
5
Dense and objective — no small talk to wade through.
Nice weather!
Haha!
POP!
What did you have for lunch?
It was amazing!
POP!
TOOL OUTPUT (DB QUERY) RESULT
order_id	total_usd
8731	129.50
8732	42.75
rows: 2
elapsed_ms: 38
status: SUCCESS
Signal, not noise. Facts, values, results.
6
So a fair test must use trajectories. That's what AMA-Bench does.
TEST MEMORY ON TRAJECTORIES, NOT CONVERSATIONS.
· States
· Actions
· Observations
· Tool Outputs
· Causal
· Objective
· Long-Horizon
THE REAL WORK LIES AHEAD


1
AMA-Bench has two halves.
REAL-WORLD
SYNTHETIC
AMA-BENCH
2
Real-world: six domains — Web, open-world QA, Text2SQL, software engineering, gaming, embodied AI.
Web
open-world QA
Text2SQL
Software Engineering
REAL-WORLD
Embodied AI
EXPERT-CHECKED
3
Real trajectories, with questions written and checked by experts.
REAL-WORLD EXAMPLE
Source: Web agent trajectory
(Task: research & summarize)
Excerpt of trajectory:
... ACTION: search("renewable energy policy 2024")
OBSERVATION: [web page html]
ACTION: click(link: /reports/2024-policy.pdf)
TOOL OUTPUT: [pdf text extracted]
ACTION: summarize(section 3)
OBSERVATION: "In 2024, the subsidy rate increased to 25%..."
...
EXPERT QUESTION:
What is the subsidy rate in 2024?
EXPERT ANSWER:
25%
EXPERT VERIFIED
4
Synthetic: rule-generated tasks in programmatic environments.
RULE ENGINE
Programmatic environment
Rules define the world, objects, actions, and rewards.
Questions are generated from ground-truth states and outcomes.
#1003
#1002
#1003
#1002
#1003
#1004
5
And synthetic tasks scale to any horizon length.
HORIZON LENGTH
SHORT
LONG
1
STATE
2
ACTION
3
4
1000
STATE
999
1001
OBSERVATION
1002
TOOL OUTPUT
1003
STATE
As long as you need.
The rules keep it consistent.
The horizon keeps growing.
6
Realistic and scalable — that's the design.
REAL-WORLD
SYNTHETIC
Two complementary halves.
One goal: measure long-horizon memory in the way agents actually work.


1
I ask four kinds of question.
TEMPORAL INFORMATION
STATE DEPENDENCY
MEMORY UPDATING
SUMMARIZATION
2
TEMPORAL INFORMATION
Temporal — what happened, and in what order?
09:00 Open config.yaml
09:10 Run backup.sh
09:22 Error in backup
09:35 Fix path in config
09:50 Backup success
It was at 09:50!
TIME
QUESTION: When did the backup succeed?
ANSWER: 09:50
3
STATE DEPENDENCY
LOCKED
State dependency — given what you did before, what's true now?
Step 1 Look around
Step 2 Pick up rusty key
Step 3 Try door
Door unlocked.
QUESTION: Can you open the door right now?
ANSWER: Yes, because I picked up the key earlier.
4
MEMORY UPDATING
Memory updating — the value changed. What is it now?
Earlier note (stale)
Retry limit = 5
Later update (new)
Retry limit = 8
updated!
QUESTION: What is the retry limit now?
ANSWER: 8
5
SUMMARIZATION
Summarization — distill the whole journey, faithfully.
Backup uploaded
SUMMARY
Goal: Back up data reliably.
Issue: Wrong path caused failure.
Fix: Corrected path in config.
Result: Backup succeeded at 09:50.
Report: Uploaded.
QUESTION: In short, what happened?
ANSWER: (See summary.)
6
Four ways for my memory to be tested... or to fail.
Each probes a different way memory can break.
AMA-BENCH: FOUR QUESTION CATEGORIES
TEMPORAL INFORMATION
STATE DEPENDENCY
MEMORY UPDATING
SUMMARIZATION
One memory. Four angles. Many chances to learn.


1
Why is the gate open now?
EXAMINATION HALL
QUESTION
(State Dependency)
Why is the gate open now?
(Because of what you did earlier.)
Let me check my memory...
AMA-BENCH
Agent Memory Evaluation
2
Similarity search grabs the note that looks alike — and misses the one that actually caused it.
FIND MOST SIMILAR NOTE
PICKED (similar)
09:48 The gate is open.
Reason: routine maintenance unlocked it.
SIMILARITY SCORE
0.92
MISSED (causal)
09:37 You used the master key to unlock the gate.
(This action changed the state.)
SIMILARITY SCORE
0.61
Looks alike (about "gate open")...
...but this is what actually caused it.
09:30 Pick up master key
09:37 Unlock gate
09:38 Gate opens
09:48 Check gate
3
Lossy compression had already thrown away the exact detail.
ORIGINAL TRAJECTORY (many details)
09:30 Pick up master key key_id = MK-73
09:31 Go to north gate key in inventory: MK-73
09:37 Unlock gate (lock_id = G-12)
09:38 Gate state = OPEN by actor: Engy
09:39 Log event {actor: Engy, gate: G-12, state: open}
··· many more steps ···
LOSSY COMPRESSION (tiny)
GATE SUMMARY
The gate was opened and checked.
Details removed:
• who opened it
• how it was opened
• which key
• lock id
• exact time
MISSING DETAIL (needed!)
Unlocked using master key MK-73 at 09:37.
4
Um... it was always open?
INCORRECT
ANSWER
The gate was always open.
?
GROUND TRUTH
The gate is open because you unlocked it earlier (at 09:37).
5
Even the best models reach only ~72%.
AMA-BENCH RESULTS (Overall Accuracy)
Frontier Model (best)
GPT-5.2
72.26%
Other Memory Systems (examples)
Baseline A (similarity RAG) 58.31%
Baseline B (summary+retrieve) 49.12%
Baseline C (agentic memory) 46.05%
Baseline D (graph RAG) 44.87%
... many others ...
Long-Context Baseline (keep everything)
Long Context (no external memory) 56.10%
On long-horizon agent tasks, many memory systems underperform plain long-context baselines.
6
The model isn't the weak link. The memory design is — it loses the causal, objective facts.
WHAT THE TASK NEEDS
Causal links (who did what, when, why)
Objective facts (values, IDs, states)
Exact details (times, IDs, parameters)
WHAT FAILING METHODS LOSE
Causal structure (flattened)
Objective detail (compressed away)
Exact facts (missed by similarity)
DIAGNOSIS
Problem: Memory method loses what matters.
If we keep the cause and the facts, we can answer correctly.


1
What if memory kept the causes, not just look-alike notes?
BEFORE (unstructured notes)
UPGRADE (organize by cause)
AMA-Agent restructures memory into a linked map of what caused what.
2
AMA-Agent builds a causality graph: action → state change → observation.
CAUSE
EFFECT
ACTION
09:37
Unlock gate (using master key)
STATE CHANGE
09:38
Gate state = OPEN (by actor: Engy)
OBSERVATION
09:39
Gate is open. (visual confirmed)
Action
State
Observation
Causal link
3
And I can use a tool to fetch the exact node — by following the cause, not the resemblance.
TOOL-AUGMENTED RETRIEVAL
QUESTION: Why is the gate open now?
APPROACH: Follow causes, then fetch the supporting facts.
Follow the causal path from the observation back to the action.
TOOLS
Graph Walk
Time Filter
Key Lookup
Fact Check
4
QUESTION: Why is the gate open now?
The gate is open because I unlocked it three steps back.
Correct.
09:39
OBSERVATION
Gate is open.
09:38
STATE
Gate state = OPEN
09:37
ACTION
Unlock gate (using master key)
CORRECT
ANSWER: I unlocked the gate at 09:37, which caused the state to become OPEN, so the gate is open now.
5
57.22% — eleven points over the best baseline.
AMA-BENCH RESULTS (Overall Accuracy)
Strongest Baseline
46.06%
AMA-Agent (ours)
57.22%
+11.16%
Other Memory Systems (examples)
• Baseline A (similarity / RAG) 49.12%
• Baseline B (summary + retrieve) 46.05%
• Baseline C (memory buffer) 44.87%
Plain long-context baseline (keep everything) 56.10%
On long-horizon agent tasks, many memory systems do worse than plain long-context baselines.
6
Keep the causes. Retrieve with tools. That's the difference.
A method that respects causality wins the test.
AMA-AGENT
CAUSALITY + TOOLS = BETTER MEMORY
ACADEMIC ADVANCE THROUGH MEANINGFUL MEMORY.


1
The lesson: it's not the size of the model. It's the design of the memory.
WHAT MAKES BETTER MEMORY?
BIGGER MODEL
SMALL GAIN
BETTER MEMORY DESIGN
BIG GAIN
IMPACT ON LONG-HORIZON TASKS
The bottleneck isn't the base model. It's the memory system we build around it.
2
Keep the causes, retrieve well — design is everything.
CAUSES (not just looks-alike)
CAUSAL LINKS (explicit and walkable)
OBJECTIVE FACTS (kept, not lost)
Good memory comes from good design choices.
3
And there's a challenge built on exactly this...
TUE STATE
TUE STATE
WED STATE
ENGRAM
A challenge for memory systems.
4
ENGRAM: read the dated states in order — never peek at tomorrow.
SAT STATE
THU STATE
FRI STATE
TODAY (WED) STATE
NO HINDSIGHT RULE
Answer using only states up to today. Don't look ahead.
QUESTION: Why is the gate open now?
ANSWER (today): It's open because I unlocked it on MON at 09:37.
5
Is this fact current? stale? superseded? uncertain? — and here's my evidence.
CURRENT
True now.
STALE
True before, not now.
SUPERSEDED
Replaced by a newer fact.
UNCERTAIN
Not enough evidence yet.
FACT UNDER REVIEW:
The gate requires a key.
JUDGMENT: CURRENT
EVIDENCE (cite):
MON 09:37 Unlocked gate (using master key MK-73) → state change recorded.
CITED TILE
MON 09:37
ACTION
Unlock gate (using MK-73)
6
Measure memory honestly.
Then build one worth measuring. That's ENGRAM.
BETTER MEMORY
BETTER AGENTS
BETTER OUTCOMES
ENGRAM
Build memory systems that respect time, causality, and facts — and answer without hindsight.
TEMPORAL
STATE DEPENDENCY
MEMORY UPDATING
SUMMARIZATION
AMA-Bench tells us what's hard.
ENGRAM invites us to build what works.